

Review Paper Notes

Why Horizontal Integration is Most Suitable

1. Improves Robustness and Generalisation

Horizontal integration combines the same omics type across multiple cohorts/datasets.

This increases:

sample diversity

population variability

dataset heterogeneity

As a result, models are less likely to overfit to one dataset and generalise better to unseen cohorts.

2. Helps Address HDLSS Problem

HDLSS = many features, few samples.

By combining cohorts:

number of patients increases

model learns more stable patterns

overfitting risk decreases

3. Enables Stronger Validation

Horizontal integration naturally supports:

cross-cohort testing

external validation

robustness evaluation

This aligns directly with my research goal.

Why Other Integration Types Are Less Suitable

Vertical Integration

Uses multiple omics from the same patients.

Limitation:

requires complete omics for every patient

many samples are dropped due to missing modalities

worsens HDLSS problem

Strong biologically, but weaker for robustness-focused research.

Mosaic Integration

Handles missing omics within one cohort.

Limitation:

mainly solves missing-data issue

still often limited to one cohort

does not strongly improve generalisation

Flexible, but not ideal for robustness across datasets.

Diagonal Integration

Combines different omics from different cohorts.

Limitation:

difficult alignment across datasets

high modelling complexity

weaker biological consistency

Too unstable and complex for my primary focus on robustness and interpretability.

Horizontal integration is most appropriate because it directly supports robustness, generalisability, and external validation by leveraging multiple cohorts while reducing HDLSS-related overfitting

To understand the biomarker: feature importance, to understand whether the biomarker is responsible for the disease, check with multiple dataset

Validation and Clinical Translation

The Translational Gap: Successful biomarker/characteristic

discovery does not guarantee clinical utility;

Population heterogeneity —not generalization to underrepresented population ,

Different integration methods are used depending on:

size of data (HDLSS or not)

noise level

interpretability needs

complexity of relationships

availability of samples

Factorization methods (PCA, NMF, autoencoders)

Used when:

data is very high-dimensional (HDLSS)

you want to reduce features

you suspect hidden biological patterns

Not used when:

you need exact gene-level explanation

dataset is already small and simple

Example:

Cancer gene expression → reduce 20,000 genes → 50 latent factors

Kernel-based methods

Used when:

relationships are non-linear

dataset is small to medium

classification or clustering is needed

Not used when:

interpretability is important

dataset is very large (too slow)

Example:

Patient similarity for disease classification

Network-based methods

Used when:

you want biological interpretation

studying gene/protein interactions

pathway analysis

Not used when:

no prior biological knowledge available

goal is only prediction accuracy

Example:

Gene regulatory networks in cancer

Correlation / covariance methods

Used when:

you want simple, interpretable relationships

exploratory analysis

Not used when:

relationships are nonlinear

data is complex multi-omics

Example:

Gene expression vs disease outcome

Probabilistic / Bayesian method, Used when:

data is noisy or missing

uncertainty is important

small sample size

Not used when:

large-scale data (too slow)

simple problems

Example:

Predicting disease risk with missing clinical data

Deep learning methods Used when:

very large datasets

complex nonlinear relationships

multi-modal data (omics + images + clinical)

Not used when:

dataset is small (HDLSS risk)

interpretability is required

Method	When used	When NOT used
Factorization	HDLSS, dimensionality reduction	need exact features
Kernel	small-medium, nonlinear patterns	large data, interpretability needed
Network	biological interpretation	no prior biology
Correlation	simple exploratory analysis	complex nonlinear data
Bayesian	uncertainty, missing data	large-scale fast models
Deep Learning	large complex datasets	small data, need interpretability

Each method is chosen based on a trade-off between data size, complexity, interpretability, and computational cost.

There is no universal best method — in multi-omics, the choice depends on whether the goal is prediction, interpretation, or handling high-dimensional noisy data.

Most interpretable methods (best → okay)

Correlation / covariance methods

Why?

very simple relationships ($A \rightarrow B$)

easy to explain biologically

Example:

“Gene A increases disease risk”

Very easy to understand

But only captures simple relationships

Network-based methods

Why?

shows gene/protein interactions

matches biological pathways

Example

Gene A → Gene B → Protein C pathway

Highly biologically meaningful

Requires prior knowledge

Factorization methods

Why?

reduces data into latent biological factors

Example:

“immune signal”, “tumor growth signal”

Some interpretability

Latent factors are not always clearly defined

Bayesian / probabilistic models

Why?

gives probability-based explanations

handles uncertainty well

interpretable in terms of risk/likelihood
mathematically complex

Low interpretability method , Kernel methods

work in “similarity space”

hard to explain features

Deep learning

black-box

needs SHAP/attention to explain

Most interpretable:

Correlation → Networks → Factorization → Bayesian → Kernel → Deep Learning (least)

For interpretability in multi-omics, simple statistical methods and biological networks are best, while deep learning and kernel methods are more powerful but less explainable. DRs = PCA, tsne, Umap, mds

Batch correction: Z-scaled, log transform, Combat, MNN, min max scaling, limma, deeplearning
Dimensionality Reduction: Proper feature selection (e.g., highly variable genes) and normalisation are critical before Dimensionality Reduction
Curse of dimensionality and sparsity.

- Omics data may have thousands of features and extreme sparsity.
- Proper feature selection (e.g., highly variable genes) and normalisation are critical before Dimensionality Reduction;
- Naive application can amplify noise.

Missing Data Imputation:

Preprocessing (log-transform, variance stabilisation, distance metric) is important before DR. To evaluate embeddings do the cross entropy test (quantitative metrics).

Clustering: k-means, hierarchical, Leiden algorithm

Classification: Random forests, gradient boosting, deep neural networks for classification

Validation: C-Index, AUC, log rank test

Workflow for clinically validated biomarkers involve: (feature selection using statistical tests(p-values, t-tests) or regularized regression (lasso, elastic net) to reduce from thousands of features to dozens.

Model Development: Train predictive models (random forests, svm, gradient boosting) on discovery cohorts with robust cross validation.

Biomarker refinement: Narrow signature to practical number of features (<20) for clinical implementation

Signature Validation: Rigorously test on independent validation cohorts; assess clinical utility (sensitivity, specificity, ROC-AUC)

DL models: Autoencoder, Graph Neural Network, Attention Based Transformer Models

Interpretable AI : SHAP, Attention Mech